

Advanced Mockito

Exercise 1: Mocking Databases and Repositories

Problem

Test a service that interacts with a database repository.

Setup

Create Repository, Service and ServiceTest classes.

Java Implementation

```
import static org.mockito.Mockito.*;
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;

public class ServiceTest{
    @Test
    public void testServiceWithMockRepository(){
        Repository repo=mock(Repository.class);
        when(repo.getData()).thenReturn("Mock Data");
        Service s=new Service(repo);
        assertEquals("Processed Mock Data",s.processData());
    }
}
```

Output

Program executes successfully and all Mockito tests pass.

Exercise 2: Mocking External Services (REST APIs)

Problem

Mock an external REST client.

Setup

Create RestClient, ApiService and test class.

Java Implementation

```
RestClient client=mock(RestClient.class);
when(client.getResponse()).thenReturn("Mock Response");
ApiService api=new ApiService(client);
assertEquals("Fetched Mock Response",api.fetchData());
```

Output

Program executes successfully and all Mockito tests pass.

Exercise 3: Mocking File I/O

Problem

Mock file reader and writer.

Setup

Create FileService with mocked FileReader/FileWriter.

Java Implementation

```
FileReader fr=mock(FileReader.class);
FileWriter fw=mock(FileWriter.class);
when(fr.read()).thenReturn("Mock File Content");
FileService fs=new FileService(fr,fw);
assertEquals("Processed Mock File Content",fs.processFile());
```

Output

Program executes successfully and all Mockito tests pass.

Exercise 4: Mocking Network Interactions

Problem

Mock network client.

Setup

Create NetworkService and NetworkClient mock.

Java Implementation

```
NetworkClient nc=mock(NetworkClient.class);
when(nc.connect()).thenReturn("Mock Connection");
NetworkService ns=new NetworkService(nc);
assertEquals("Connected to Mock Connection",ns.connectToServer());
```

Output

Program executes successfully and all Mockito tests pass.

Exercise 5: Mocking Multiple Return Values

Problem

Return different values on consecutive calls.

Setup

Use thenReturn() chaining.

Java Implementation

```
Repository repo=mock(Repository.class);  
when(repo.getData())  
    .thenReturn("First Mock Data")  
    .thenReturn("Second Mock Data");  
Service s=new Service(repo);  
assertEquals("Processed First Mock Data",s.processData());  
assertEquals("Processed Second Mock Data",s.processData());
```

Output

Program executes successfully and all Mockito tests pass.